ATOMIC-LEVEL MATERIALS SIMULATION

Module Code: MES306TC

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Spin Orbit Coupling

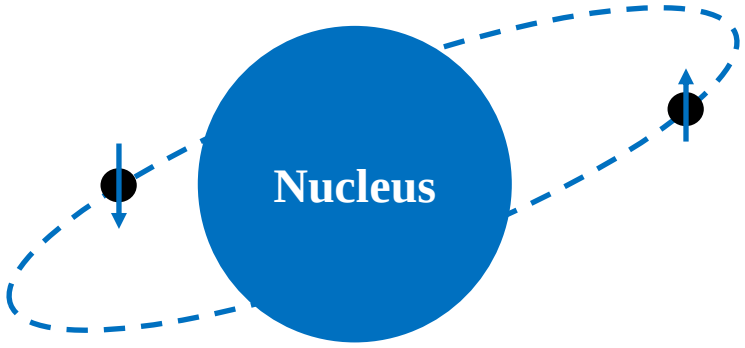
I. SOC Formalism: Intrinsic, Rashba and Dresselhaus

II. Spin Texture and computing procedure

III. Visualization: The Physicality of Spin-Momentum Locking Feature

I. SOC Formalism: Intrinsic, Rashba and Dresselhaus

Intrinsic Spin Orbit Coupling (SOC)



***p* Shell Orbital Electron**

	$\|1,\uparrow\rangle$	$\|-1,\downarrow\rangle$	$\|1,\downarrow\rangle$	$\|0,\uparrow\rangle$	$\|0,\downarrow\rangle$	$\|-1,\uparrow\rangle$
$\|1,\uparrow\rangle$	1/2	0	0	0	0	0
$\|-1,\downarrow\rangle$	0	1/2	0	0	0	0
$\|1,\downarrow\rangle$	0	0	-1/2	$1/\sqrt{2}$	0	0
$\|0,\uparrow\rangle$	0	0	$1/\sqrt{2}$	0	0	0
$\|0,\downarrow\rangle$	0	0	0	0	0	$1/\sqrt{2}$
$\|-1,\uparrow\rangle$	0	0	0	0	$1/\sqrt{2}$	-1/2

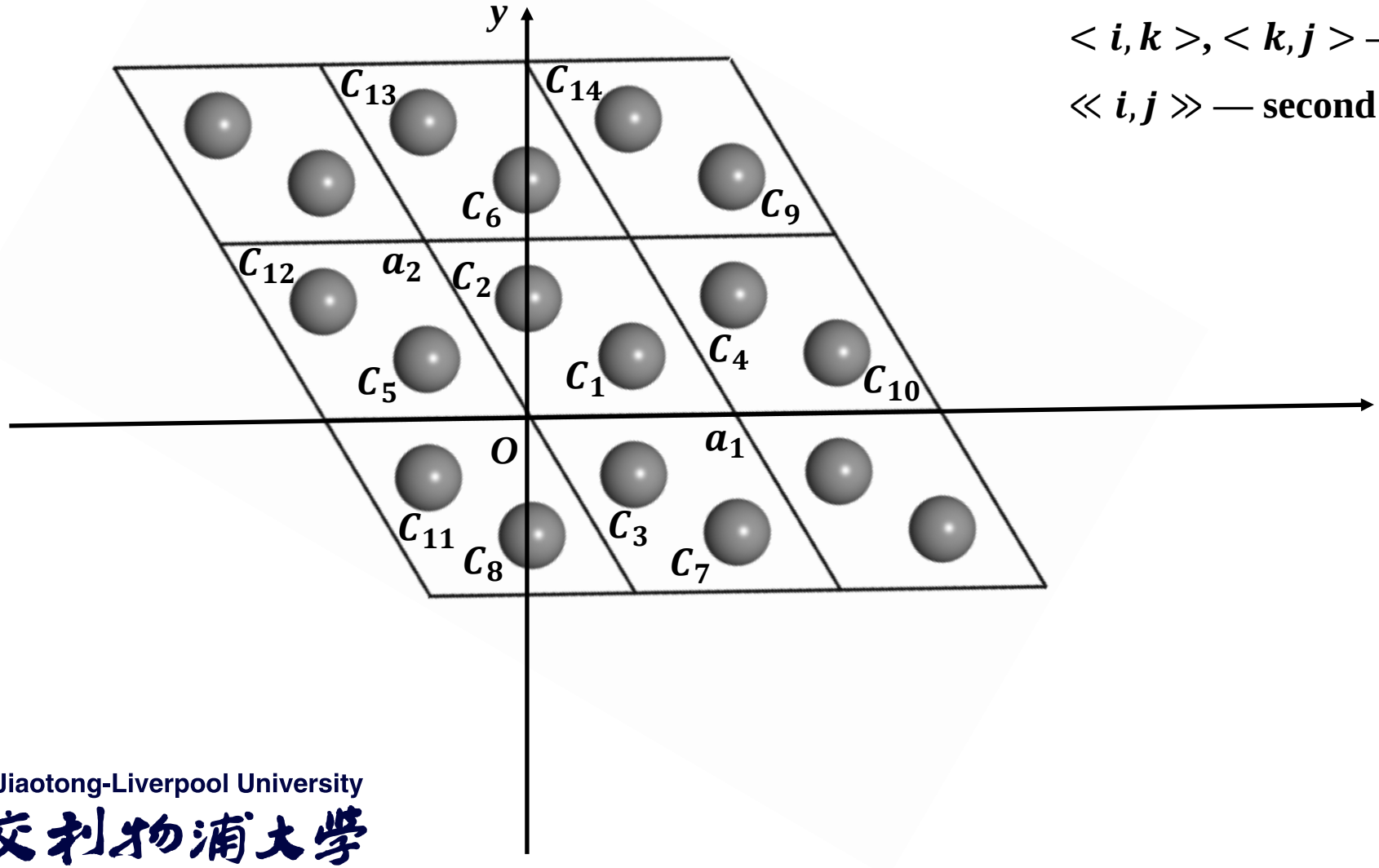
$$\langle m_l, m_s | L \cdot S | m_l, m_s \rangle =$$

- ♦ A 4-fold degenerate state with E = 1/2 (Total angular momentum j = 3/2)
- ♦ A 2-fold degenerate state with E = -1 (Total angular momentum j = 1/2)

Spin and orbital momentum could couple together and bring the energy difference. Physically, it flips the spin direction

I. SOC Formalism: Intrinsic, Rashba and Dresselhaus

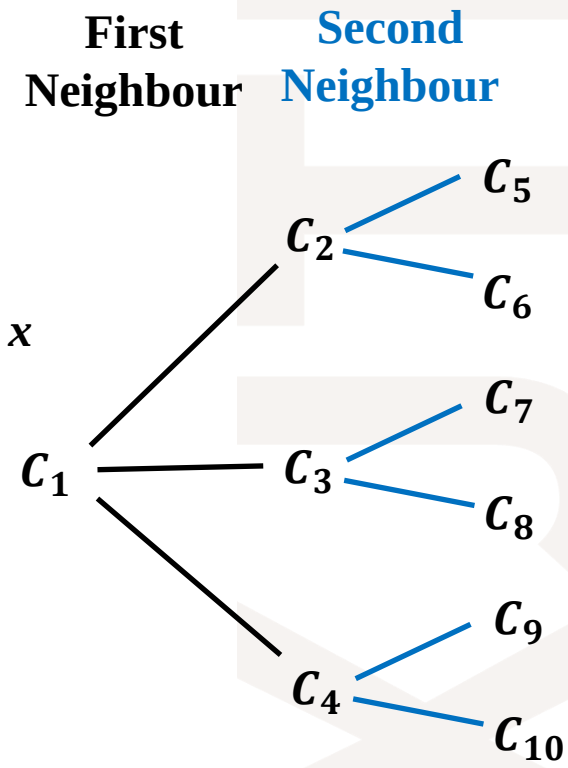
Structure of Graphene



$$H_{intrinsic_soc} = it_{intrinsic_soc} \sum_{\ll i,j \gg} c_i^{\dagger} \vec{u}_{ik} \times \vec{u}_{kj} \sigma_z c_j$$

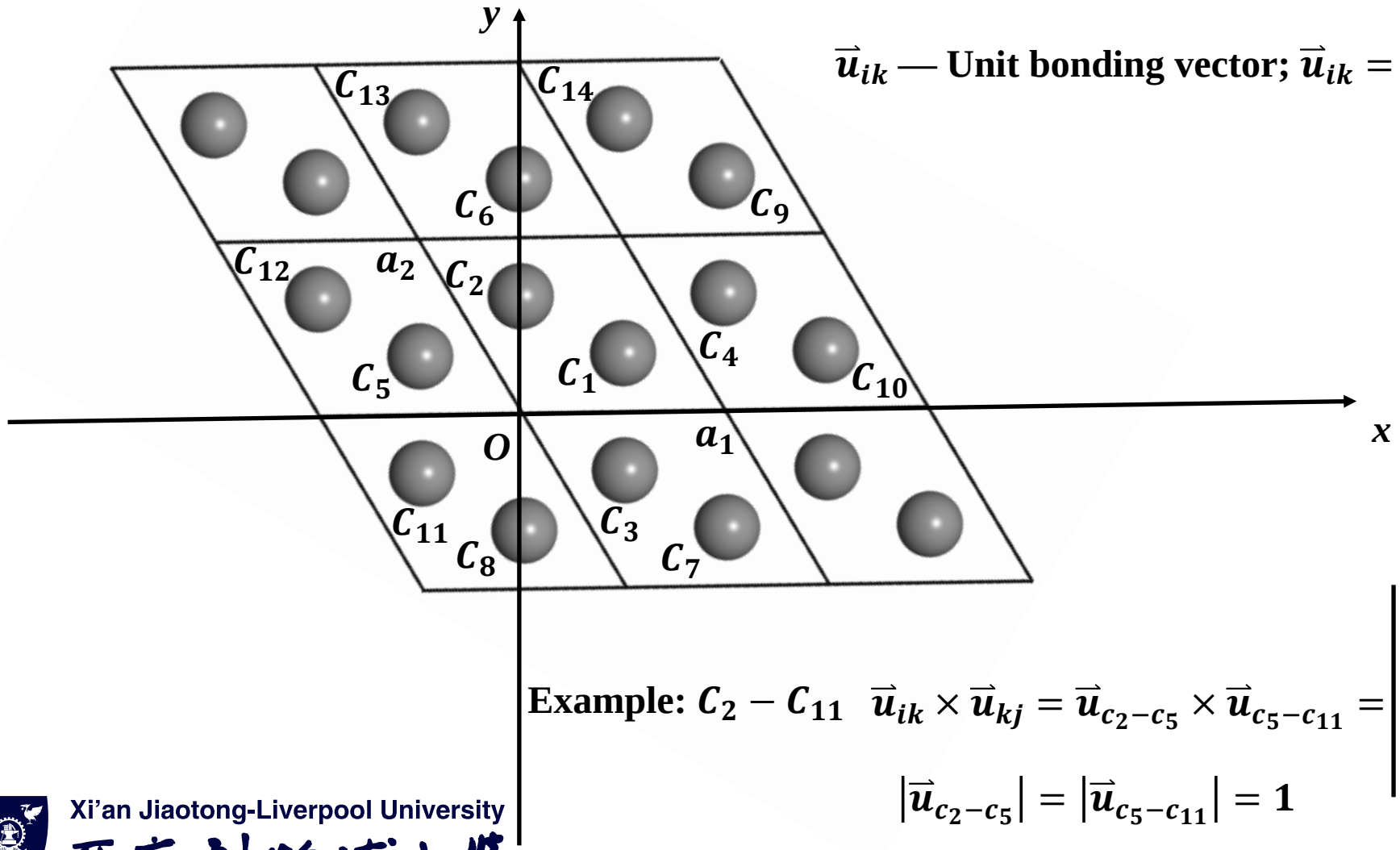
$< i, k >, < k, j >$ — first neighbour atom pair

$\ll i, j \gg$ — second neighbour atom pair



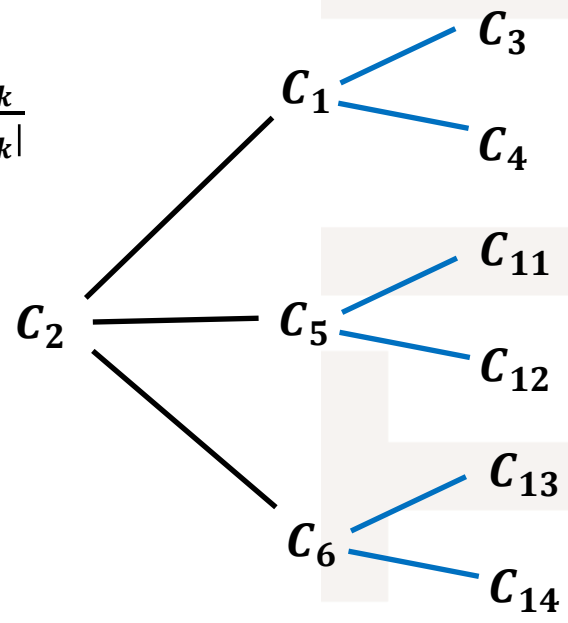
I. SOC Formalism: Intrinsic, Rashba and Dresselhaus

Structure of Graphene



$\vec{u}_{ik}$ — Unit bonding vector; $\vec{u}_{ik} = \frac{\vec{u}_{ik}}{|\vec{u}_{ik}|}$

First Neighbour **Second Neighbour**



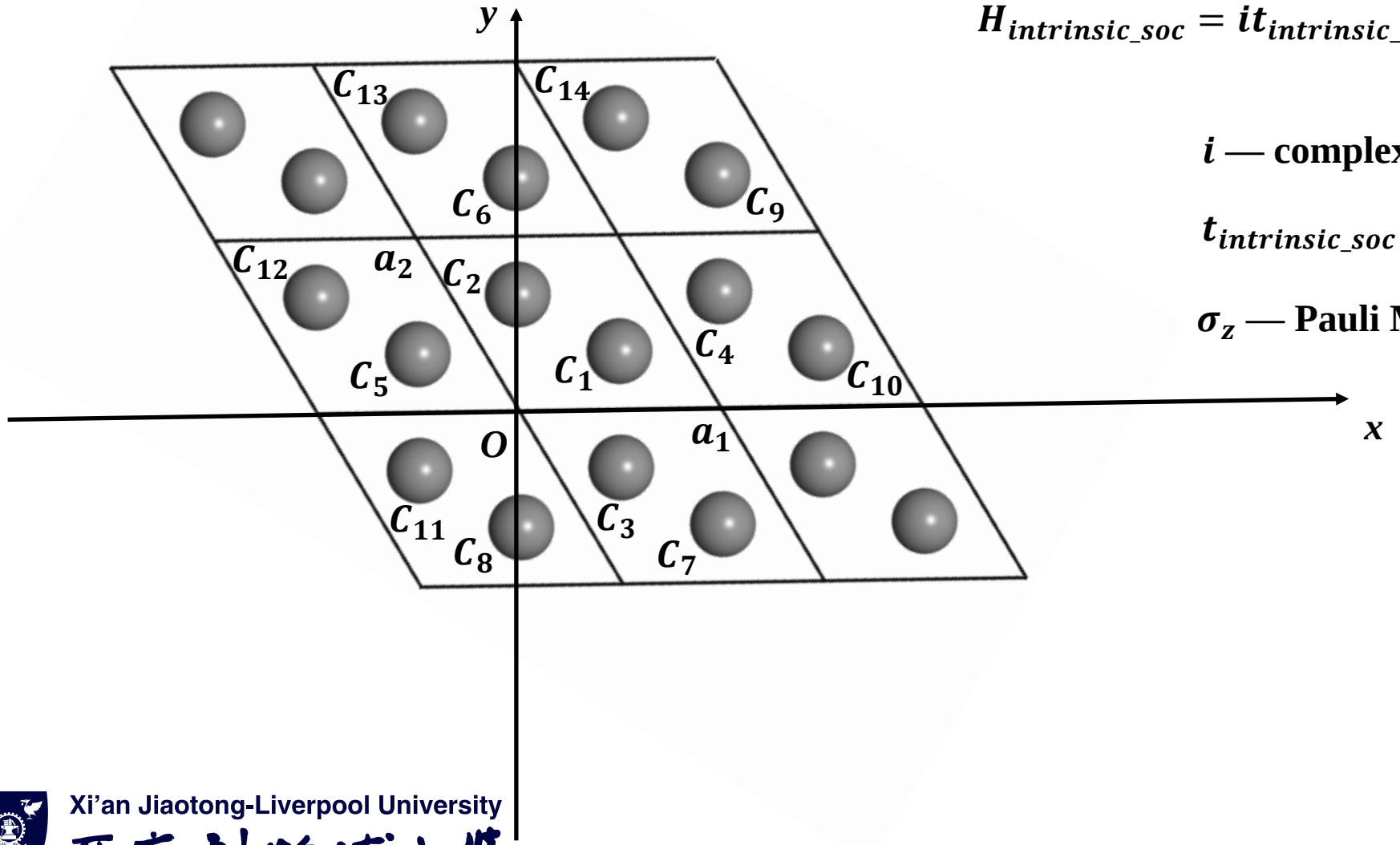
Example: $C_2 - C_{11}$ $\quad \vec{u}_{ik} \times \vec{u}_{kj} = \vec{u}_{c_2-c_5} \times \vec{u}_{c_5-c_{11}} =$

$$\begin{vmatrix} x & y & z \\ \vec{u}_{c_2-c_5}(x) & \vec{u}_{c_2-c_5}(y) & \vec{u}_{c_2-c_5}(z) \\ \vec{u}_{c_5-c_{11}}(x) & \vec{u}_{c_5-c_{11}}(y) & \vec{u}_{c_5-c_{11}}(z) \end{vmatrix}$$

$$|\vec{u}_{c_2-c_5}| = |\vec{u}_{c_5-c_{11}}| = 1$$

I. SOC Formalism: Intrinsic, Rashba and Dresselhaus

Structure of Graphene



$$H_{intrinsic_soc} = it_{intrinsic_soc} \sum_{\ll i,j \gg} c_i^{\dagger} \vec{u}_{ik} \times \vec{u}_{kj} \sigma_z c_j$$

i — complex constant i

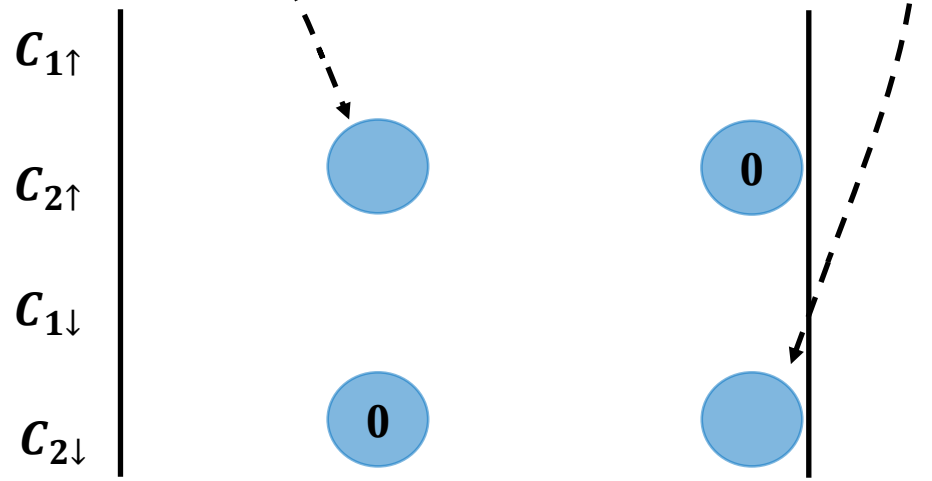
$t_{intrinsic_soc}$ — strength of intrinsic SOC term

σ_z — Pauli Matrix along z axis $\quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

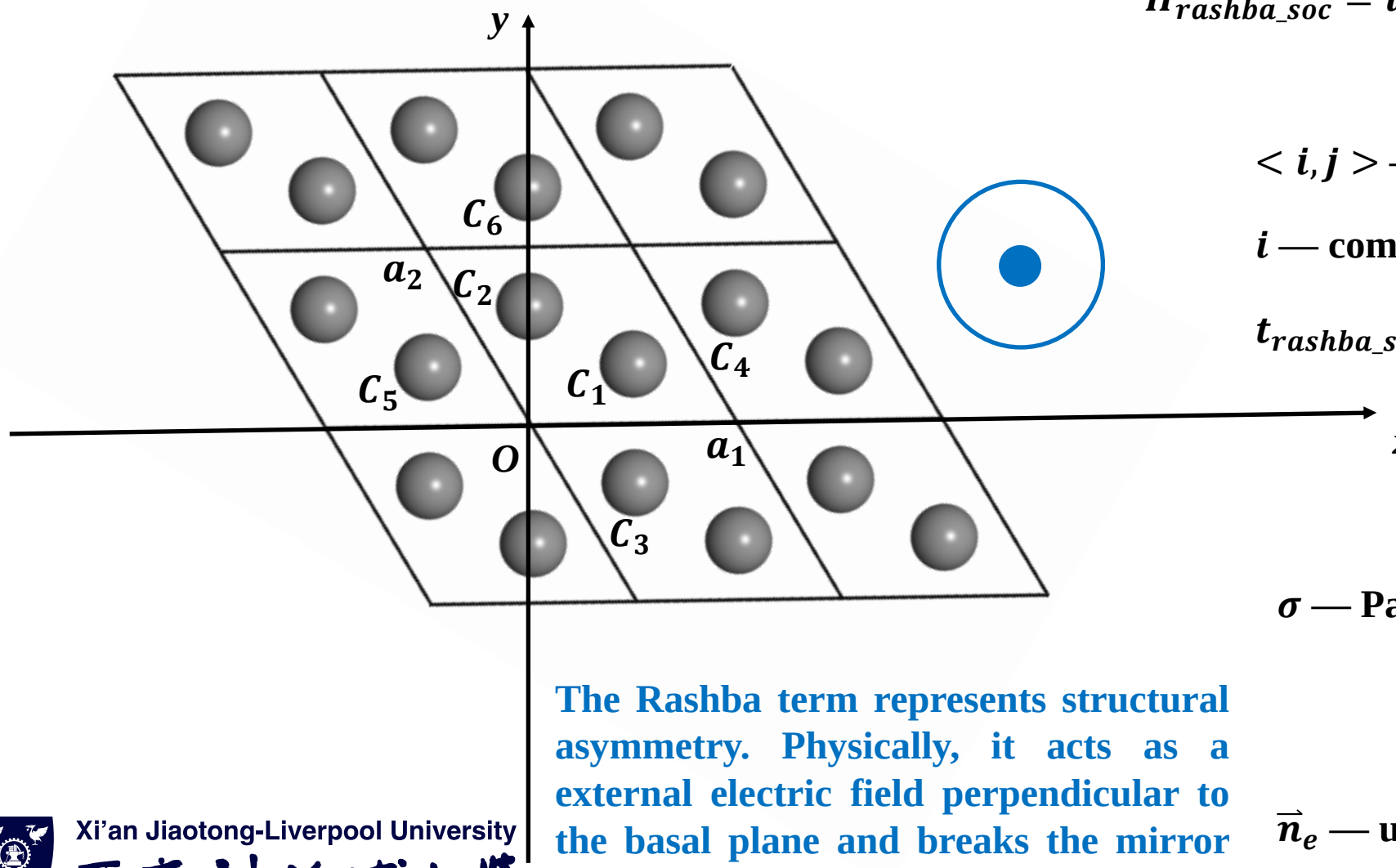
I. SOC Formalism: Intrinsic, Rashba and Dresselhaus

$$H_{intrinsic_soc} = \vec{u}_{c_2-c_5} \times \vec{u}_{c_5-c_{11}} \cdot \sigma_z = i \cdot t_{intrinsic_soc} \cdot \begin{pmatrix} \vec{u}_{c_2-c_5} \times \vec{u}_{c_5-c_{11}} & 0 \\ 0 & -\vec{u}_{c_2-c_5} \times \vec{u}_{c_5-c_{11}} \end{pmatrix} \cdot e^{i \cdot k \cdot (-1 \cdot a_1 - 1 \cdot a_2)}$$

Example: $C_2 - C_{11}$ $H_{intrinsic_soc} =$



I. SOC Formalism: Intrinsic, Rashba and Dresselhaus



$$H_{rashba_soc} = it_{rashba_soc} \sum_{<i,j>} c_i^{\dagger} \boldsymbol{\sigma} \cdot (\vec{u}_{ij} \times \vec{n}_e) c_j$$

$< i, j >$ — first neighbour atom pair

i — complex constant i

t_{rashba_soc} — strength of rashba SOC term

$\boldsymbol{\sigma}$ — Pauli Matrix

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$\sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

$$\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

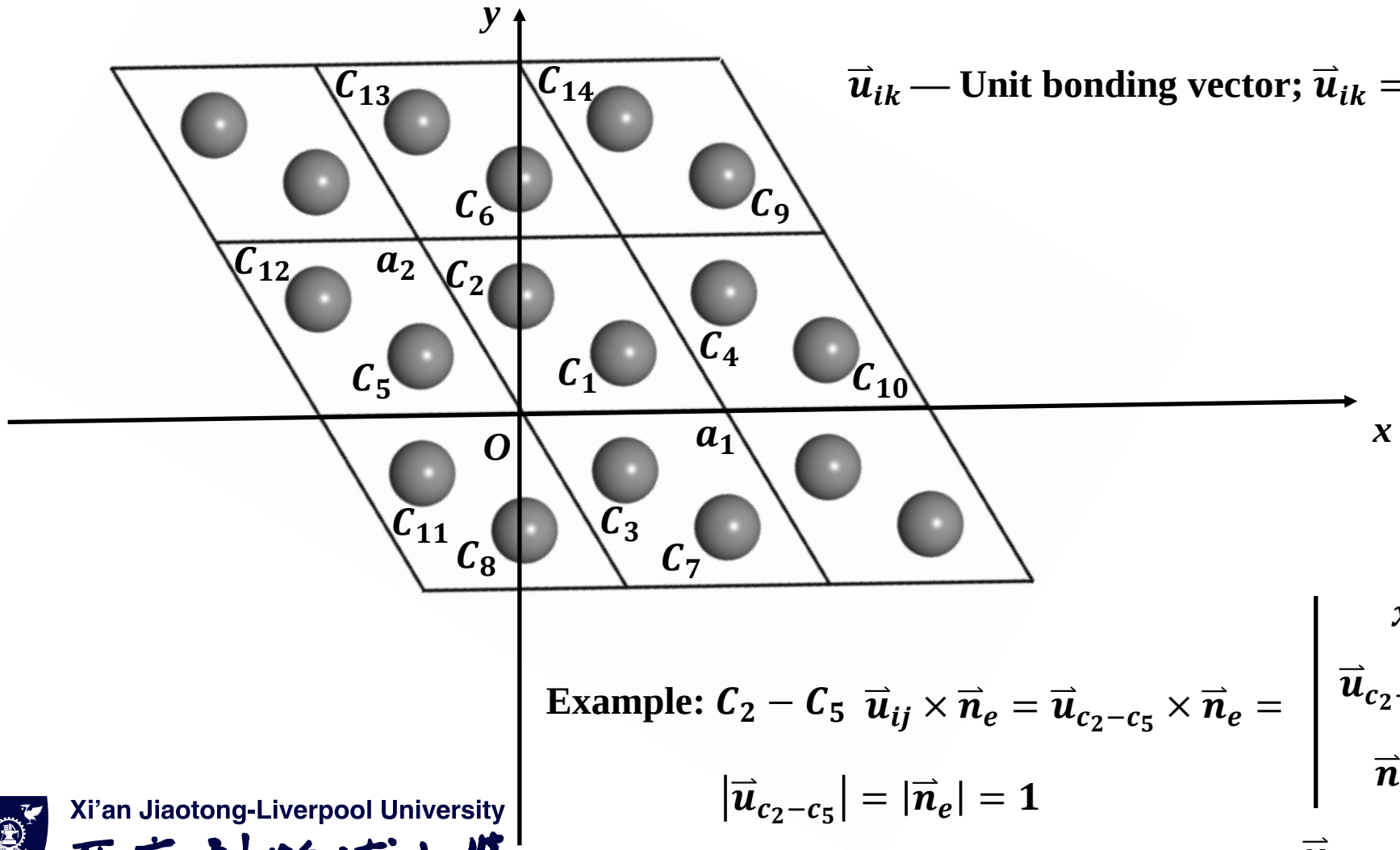
$\vec{n}_e$ — unit vector along normal direction

$|\vec{n}_e| = 1$

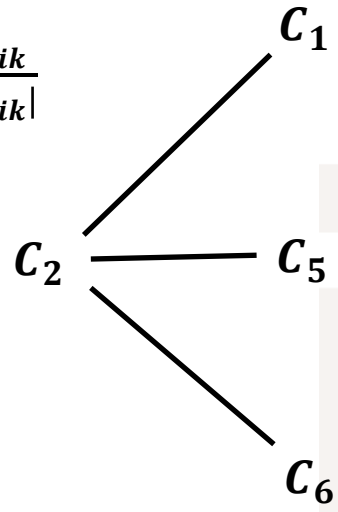
The Rashba term represents structural asymmetry. Physically, it acts as a external electric field perpendicular to the basal plane and breaks the mirror symmetry ($z \rightarrow -z$)

I. SOC Formalism: Intrinsic, Rashba and Dresselhaus

Structure of Graphene



First Neighbour



$\vec{u}_{ik}$ — Unit bonding vector; $\vec{u}_{ik} = \frac{\vec{u}_{ik}}{|\vec{u}_{ik}|}$

Example: $C_2 - C_5$ $\vec{u}_{ij} \times \vec{n}_e = \vec{u}_{c_2-c_5} \times \vec{n}_e =$

$$\begin{vmatrix} x & y & z \\ \vec{u}_{c_2-c_5}(x) & \vec{u}_{c_2-c_5}(y) & \vec{u}_{c_2-c_5}(z) \\ \vec{n}_e(x) & \vec{n}_e(y) & \vec{n}_e(z) \end{vmatrix}$$

$$= \vec{u}_{c_2-c_5}(y)_(x) - \vec{u}_{c_2-c_5}(x)_(y)$$

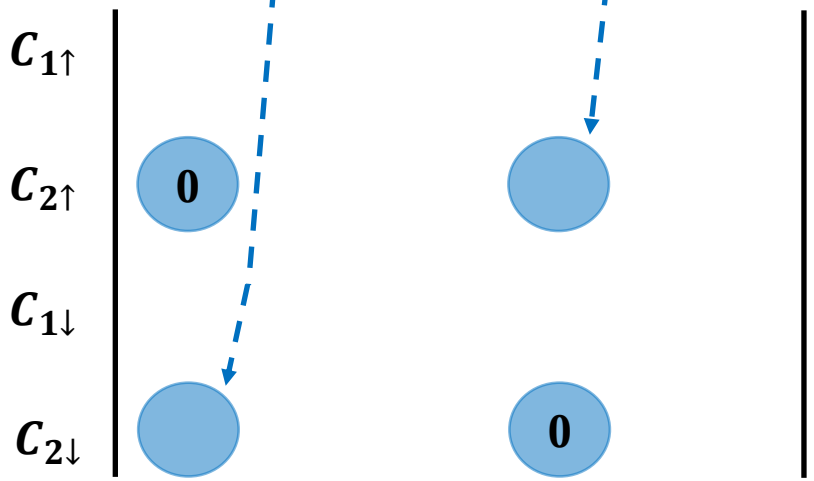
$$|\vec{u}_{c_2-c_5}| = |\vec{n}_e| = 1$$

I. SOC Formalism: Intrinsic, Rashba and Dresselhaus

$$H_{rashba_soc} = i \cdot t_{rashba_soc} \cdot \sigma \cdot (\vec{u}_{c_2-c_5} \times \vec{n}_e) = \left[\sigma_x \left(\vec{u}_{c_2-c_5}(y)\right)_x - \sigma_y \left(\vec{u}_{c_2-c_5}(x)\right)_y\right] \cdot e^{i\cdot k\cdot(-1\cdot a_1+0\cdot a_2)}$$

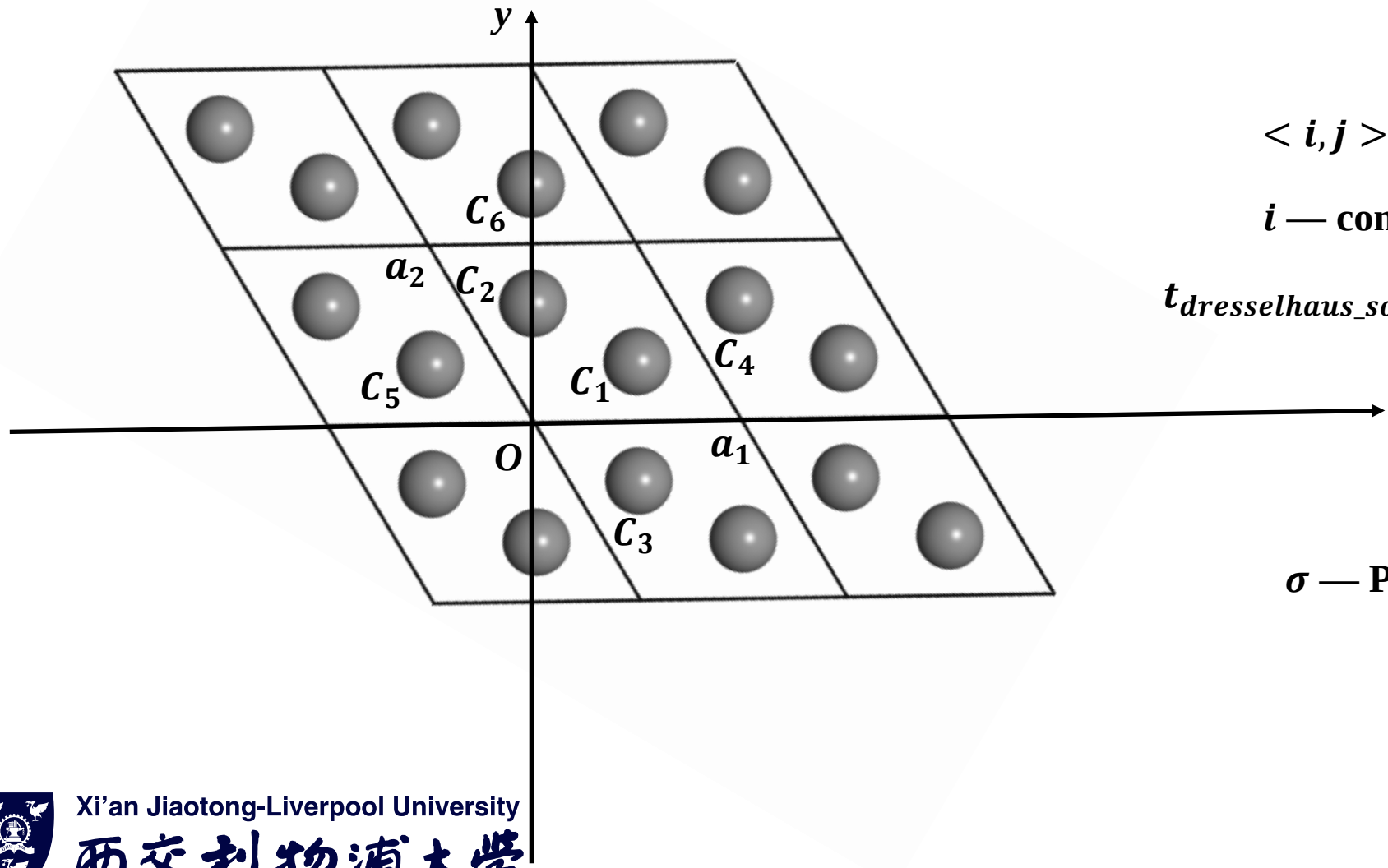
$$= \begin{pmatrix} 0 & \vec{u}_{c_2-c_5}(y) \\ \vec{u}_{c_2-c_5}(y) & 0 \end{pmatrix} - \begin{pmatrix} 0 & -i\cdot\vec{u}_{c_2-c_5}(x) \\ i\cdot\vec{u}_{c_2-c_5}(x) & 0 \end{pmatrix} \cdot e^{i\cdot k\cdot(-1\cdot a_1+0\cdot a_2)}$$

Example: $C_2 - C_5$ $H_{rashba_soc} =$



I. SOC Formalism: Intrinsic, Rashba and Dresselhaus

$$H_{dresselhaus_soc} = it_{dresselhaus_soc} \sum_{<i,j>} c_i^{\dagger}\left[\sigma_x \cdot \vec{u}_{ij}(x) - \sigma_y \cdot \vec{u}_{ij}(y)\right]c_j$$



$< i, j >$ — first neighbour atom pair

i — complex constant i

$t_{dresselhaus_soc}$ — strength of dresselhaus SOC term

σ — Pauli Matrix

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$\sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

$$\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

I. SOC Formalism: Intrinsic, Rashba and Dresselhaus

Dresselhaus SOC acts as an intrinsic mechanism that couples spin degrees of freedom to the lattice geometry, thereby imposing anisotropy on the spin-momentum locking

Lifting Spin Degeneracy: Similar to Rashba SOC, it breaks time-reversal protected spin degeneracy in materials lacking bulk inversion symmetry (BIA), splitting the energy bands even in the absence of an external magnetic field

Imposing Structural Anisotropy: Unlike the isotropic vortex-like texture produced by Rashba SOC, Dresselhaus SOC is tied to the crystal's internal axes. It introduces a direction-dependent (anisotropic) constraint on spin orientation, which deforms the spin texture based on the material's specific point group symmetry

Engineering Long-lived Spin States: When combined with Rashba SOC, Dresselhaus SOC can be used to engineer the Persistent Spin Helix (PSH) state. By tuning the Rashba strength (t_{rashba_soc}) to match the Dresselhaus strength ($t_{dresselhaus_soc}$), the spin-orbit effective magnetic fields effectively cancel each other out in specific directions, leading to a suppression of spin relaxation and significantly longer spin coherence lengths

Material Symmetry Probe: It serves as a precise spectroscopic probe for structural characterization. Since its strength is determined by the bulk crystal properties, measuring the Dresselhaus coefficient ($t_{dresselhaus_soc}$) allows researchers to identify hidden structural asymmetries or determine the crystal growth orientation

I. SOC Formalism: Intrinsic, Rashba and Dresselhaus

$$H_{dresselhaus_soc} = i \cdot t_{dresselhaus_soc} \cdot \left[\sigma_x \cdot \vec{u}_{ij}(x) - \sigma_y \cdot \vec{u}_{ij}(y)\right] \cdot e^{i\cdot k\cdot(-1\cdot a_1 + 0\cdot a_2)}$$

$$= \begin{pmatrix} 0 & \vec{u}_{c_2-c_5}(x) \\ \vec{u}_{c_2-c_5}(x) & 0 \end{pmatrix} - \begin{pmatrix} 0 & -i\cdot\vec{u}_{c_2-c_5}(y) \\ i\cdot\vec{u}_{c_2-c_5}(y) & 0 \end{pmatrix} \cdot e^{i\cdot k\cdot(-1\cdot a_1 + 0\cdot a_2)}$$

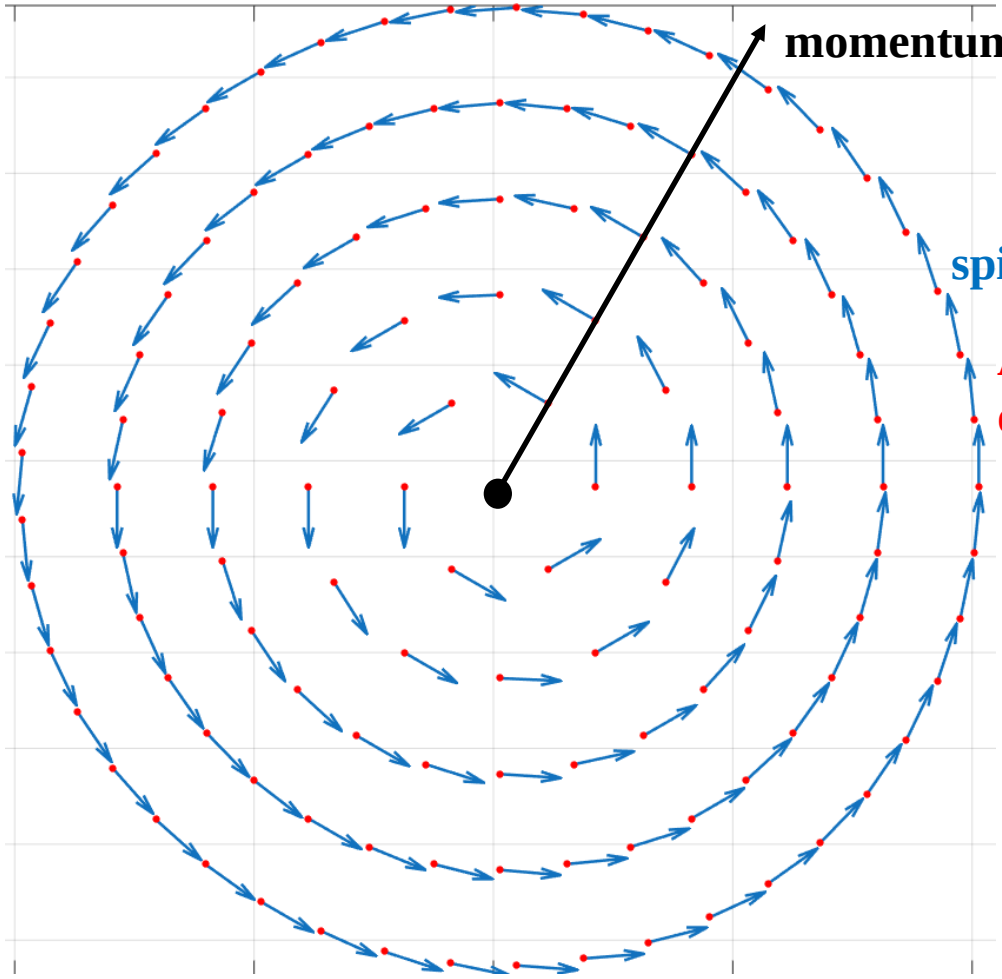
Example: $C_2 - C_5$ $H_{dresselhaus_soc} =$

	$C_{1\uparrow}$	$C_{2\uparrow}$	$C_{1\downarrow}$	$C_{2\downarrow}$
$C_{1\uparrow}$		0		
$C_{2\uparrow}$				
$C_{1\downarrow}$				0
$C_{2\downarrow}$			0	

II. Spin Texture and computing procedure

Rashba Type Texture — Vortex/Helical structure in *k* space

Rashba SOC creates a vortex. As moving around the *k* point, the spin rotates in-plane. This is the 'Rashba texture'



momentum direction — electron moving direction

↓

Charge Current Direction

spin direction

Along a certain current direction, spin direction on all electrons are same

↓

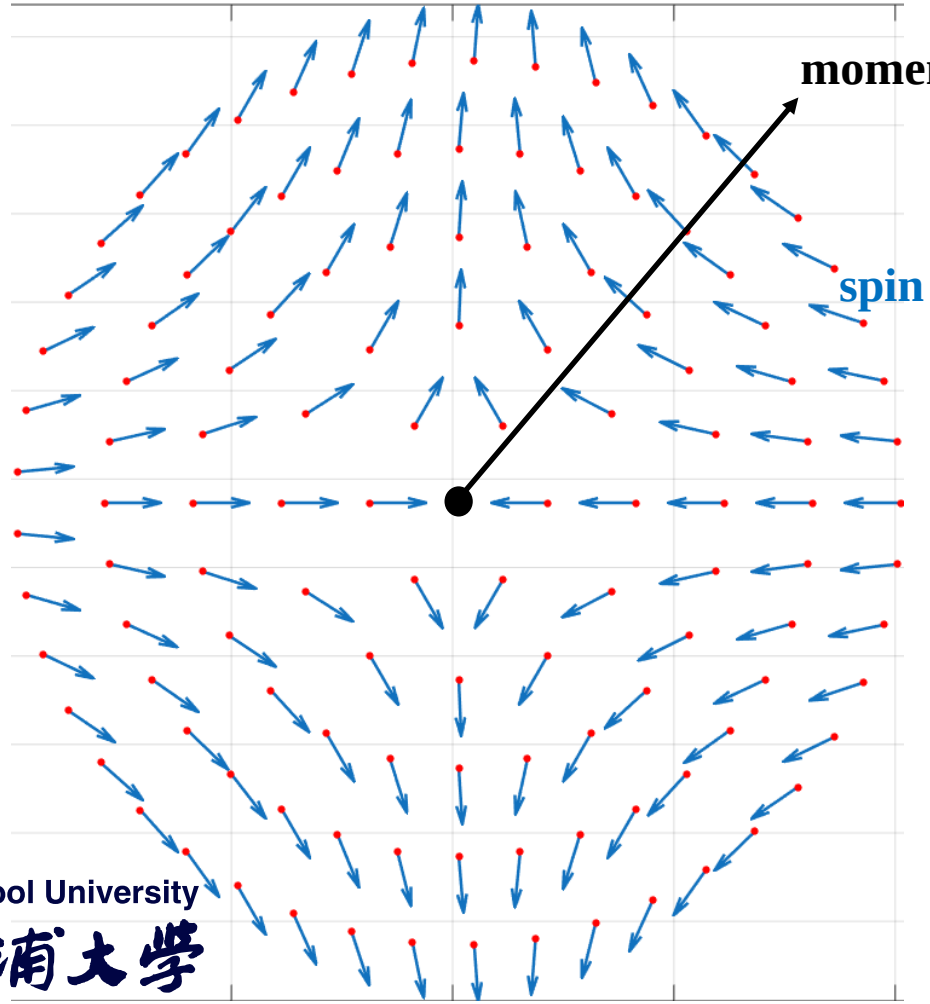
Spin-Momentum Locking

Feature of Topological Insulator

II. Spin Texture and computing procedure

Dresselhaus Type Texture — Anisotropic, symmetry-constrained rotation

Dresselhaus SOC following the crystal lattice axes, breaks the 'perfect' rotational symmetry of the Rashba vortex. It's vital to distinguish these to understand if your device is interface-dominated or bulk-dominated



momentum direction — electron moving direction

↓

Charge Current Direction

spin direction

Along a certain current direction, spin direction on all electrons are same

↓

Spin-Momentum Locking

Feature of Topological Insulator

II. Spin Texture and computing procedure

Computing Procedure:

1. Build up *k* point mesh, usually in the circle shape

2. Compose the Hamiltonian matrix: $Hamiltonian = H_{onsite} + H_{hopping} + H_{intrinsic_soc} + H_{rashba_soc}/H_{dresselhaus_soc}$

3. Diagonalise the Hamiltonian to obtain the eigenvector matrix

4. Extract the eigenvector the band to be examined, be careful with the spinor components

5. Calculate the expectation value of the spin operator

6. Take the real term of the expectation value because the imaginary term is zero

II. Spin Texture and computing procedure

Spin texture — the expectation value of the spin operator, Pauli matrice $\langle S\rangle = \langle\Psi|\sigma|\Psi\rangle$

$$\begin{cases} \langle S\rangle_x = \langle\Psi|\sigma_x|\Psi\rangle \\ \langle S\rangle_y = \langle\Psi|\sigma_y|\Psi\rangle \\ \langle S\rangle_z = \langle\Psi|\sigma_z|\Psi\rangle \end{cases}$$

$$Hamiltonian = \begin{array}{c|cccc} & C_{1\uparrow} & C_{2\uparrow} & C_{1\downarrow} & C_{2\downarrow} \\ \hline C_{1\uparrow} & & & & \\ C_{2\uparrow} & & & & \\ C_{1\downarrow} & & & & \\ C_{2\downarrow} & & & & \end{array}$$

$$Eigenvector = \begin{array}{c|cccc} & C_{1\uparrow} & C_{2\uparrow} & C_{1\downarrow} & C_{2\downarrow} \\ \hline C_{1\uparrow} & \bullet & & & \begin{pmatrix} C_{1\uparrow} \\ C_{1\downarrow} \end{pmatrix} \\ C_{2\uparrow} & & \bullet & & \\ C_{1\downarrow} & \bullet & & & \\ C_{2\downarrow} & & \bullet & & \begin{pmatrix} C_{2\uparrow} \\ C_{2\downarrow} \end{pmatrix} \end{array}$$

After solving for the eigenvector $\Psi_n(k)$**, the spin operator is projected onto the state to determine the spin orientation at that specific momentum**

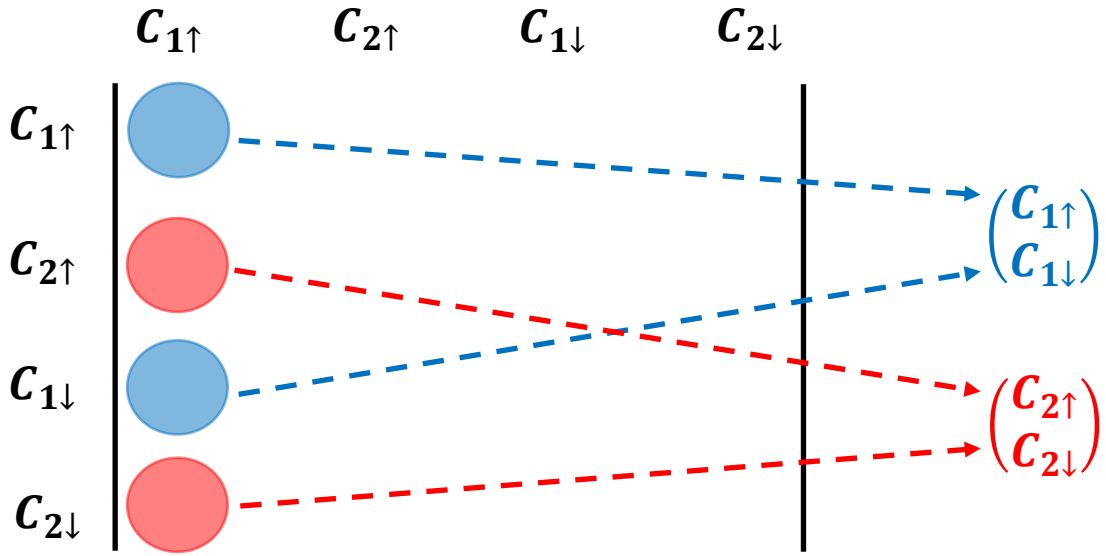
The spin is a property of the total wavefunction, which requires the sum of all components

II. Spin Texture and computing procedure

Spin texture — the expectation value of the spin operator, Pauli matrice $\langle S\rangle = \langle\Psi|\sigma|\Psi\rangle$

$$\begin{cases}\langle S\rangle_x = \langle\Psi|\sigma_x|\Psi\rangle \\ \langle S\rangle_y = \langle\Psi|\sigma_y|\Psi\rangle \\ \langle S\rangle_z = \langle\Psi|\sigma_z|\Psi\rangle\end{cases}$$

$Eigenvector =$



Example: Spin Texture on the 1st band

$$\langle S\rangle_x = \langle\Psi|\sigma_x|\Psi\rangle = \begin{pmatrix} C_{1\uparrow}^\dagger & C_{1\downarrow}^\dagger \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} C_{1\uparrow} \\ C_{1\downarrow} \end{pmatrix} + \begin{pmatrix} C_{2\uparrow}^\dagger & C_{2\downarrow}^\dagger \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} C_{2\uparrow} \\ C_{2\downarrow} \end{pmatrix}$$

$$\langle S\rangle_y = \langle\Psi|\sigma_y|\Psi\rangle = \begin{pmatrix} C_{1\uparrow}^\dagger & C_{1\downarrow}^\dagger \end{pmatrix}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}\begin{pmatrix} C_{1\uparrow} \\ C_{1\downarrow} \end{pmatrix} + \begin{pmatrix} C_{2\uparrow}^\dagger & C_{2\downarrow}^\dagger \end{pmatrix}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}\begin{pmatrix} C_{2\uparrow} \\ C_{2\downarrow} \end{pmatrix}$$

$$\langle S\rangle_z = \langle\Psi|\sigma_z|\Psi\rangle = \begin{pmatrix} C_{1\uparrow}^\dagger & C_{1\downarrow}^\dagger \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} C_{1\uparrow} \\ C_{1\downarrow} \end{pmatrix} + \begin{pmatrix} C_{2\uparrow}^\dagger & C_{2\downarrow}^\dagger \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} C_{2\uparrow} \\ C_{2\downarrow} \end{pmatrix}$$

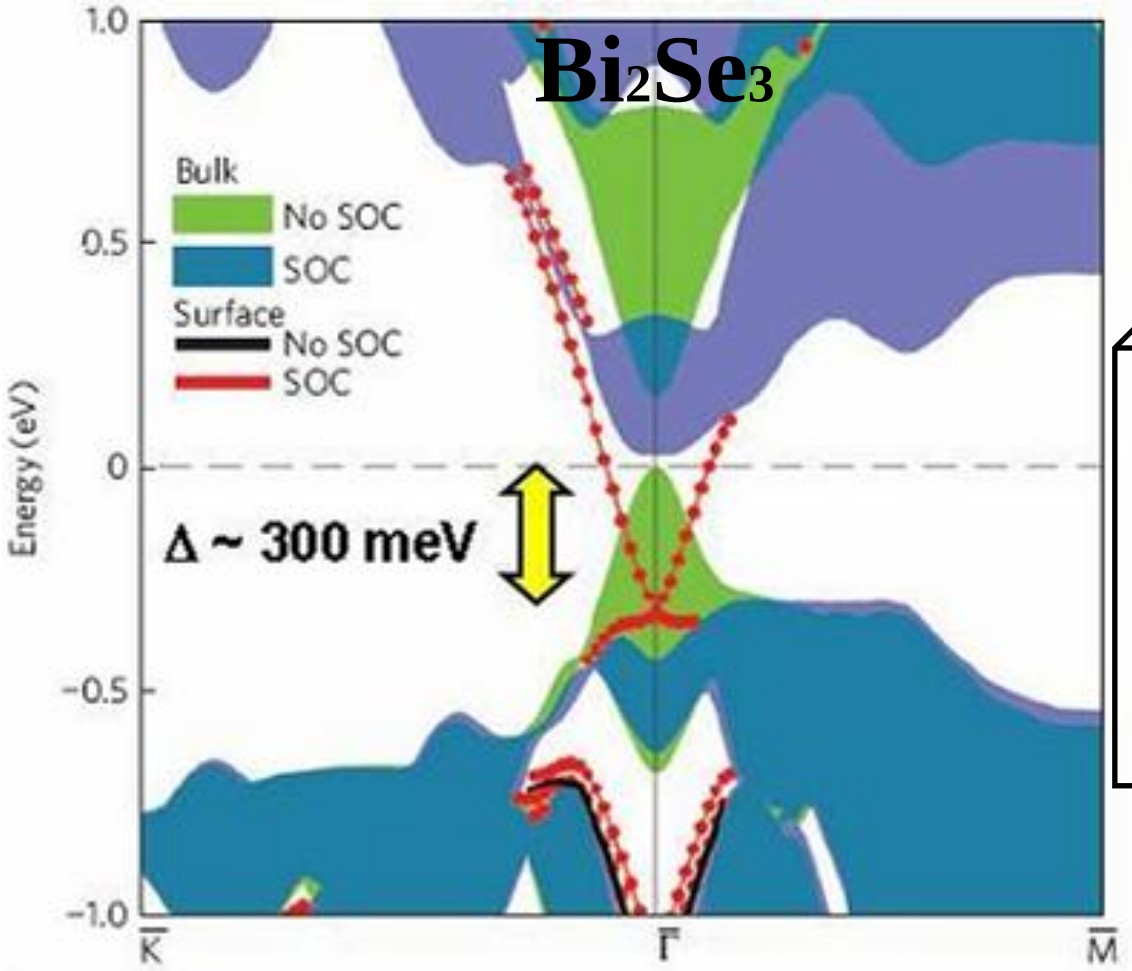
III. Visualization: The Physicality of Spin-Momentum Locking Feature

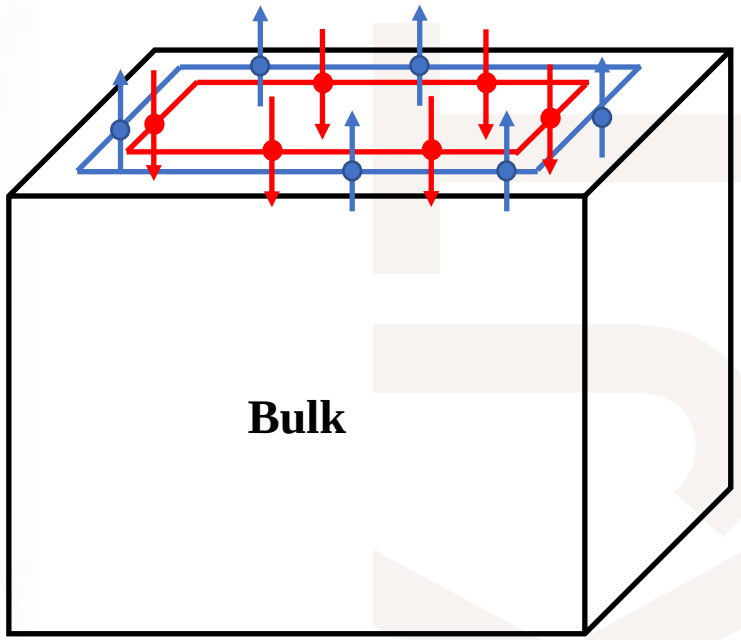
Rashba Type Texture — If a spin is locked to momentum, it cannot be backscattered by non-magnetic impurities. This is why these systems conduct electricity so perfectly

Topological Insulator — Conductor on the surface; while, insulator inside the bulk; *e.g.* Bi_2Se_3, Sb_2Se_3, Bi_2Te_3, Sb_2Te_3

The fact that the spin rotates as electrons traverse the Brillouin zone is the first hint of topology

If the spin wraps around the BZ, the winding number is non-zero. That number is the topological invariant





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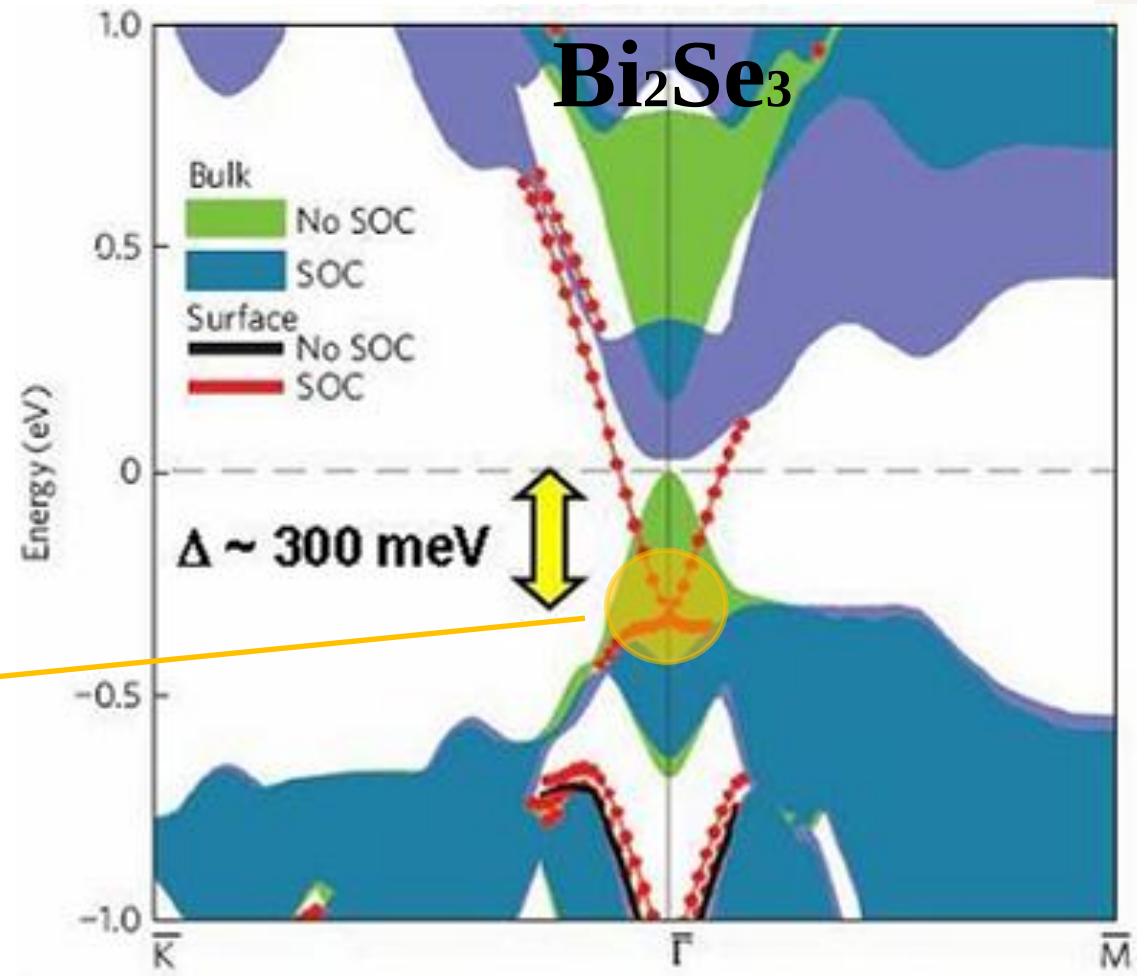
III. Visualization: The Physicality of Spin-Momentum Locking Feature

Intrinsic: $t_{intrinsic_soc}$ (Bulk property) opens a gap

Rashba: t_{rashba_soc} (Surface property) tilts the spin

Z_2 classification

- **SOC = Splitting**
- **Texture = Chirality**
- **Chirality = Topological Potential**



These are where the Berry Curvature peaks. Topology lives at these singular points



That's all for this session

Thank you for your attention and great participation